

PAVAN KUMAR MAGANDI

Data Analytics • Financial Analysis • Risk Management • Business Intelligence

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PROFESSIONAL SUMMARY

Analytics professional with an MBA in Finance and 2.5+ years of experience in data analysis, MIS reporting, and business intelligence. Built an end-to-end Global Sovereign Risk Dashboard using IMF WEO and World Bank data across 183 economies, leveraging Excel, Python, PostgreSQL, and Tableau for data cleaning, automated data integration, sovereign risk scoring, regulatory reporting, and investment decision support. Proficient in Power BI and R for dashboard development and statistical analysis.

CORE COMPETENCIES

Languages & Tools: Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn); PostgreSQL, MySQL; SQL (Joins, CTEs, Window Functions — LAG/LEAD/RANK, Subqueries, Aggregations); Advanced Excel (Power Query, ETL, XLOOKUP, INDEX-MATCH, Pivot Tables, Dynamic Arrays, Dashboard Development), Power BI, Tableau, R.

Analytics: EDA, Statistical Modelling, Financial Modelling, Data Cleaning, Data Validation, Time-Series Analysis & Forecasting, Business Intelligence, KPI & Performance Reporting, Data Visualisation.

Risk & Finance: Scenario Analysis, Hypothesis Testing, Commodity Risk, Portfolio Optimisation, ESG Analysis, Credit Risk, Sovereign & Macroeconomic Risk, Data Governance, AML & Transaction Monitoring, Country Risk Assessment, Regulatory Reporting & Compliance, Stress Testing.

KEY PROJECTS

Global Sovereign Risk Dashboard (IMF & World Bank Data) (*PostgreSQL, Tableau, Excel, Python*): Engineered an end-to-end sovereign risk early-warning system across 183 countries (2018–2026) by integrating IMF WEO and World Bank GEM datasets, deduplicating 23,149 raw records into 1,627 verified country-year observations and identifying & resolving 5 distinct data-integrity defects (including a ratio-of-averages miscalculation, null-to-zero coercion errors, and outlier distortion from anomalous readings such as a 65,374% single-year inflation spike).

- Designed a 4-pillar, 8-indicator weighted “Safe Score” composite risk model (Fiscal, External, Growth, Monetary) mirroring S&P/Moody’s/Fitch sovereign rating methodology, with pillar weights calibrated by economic significance; backtested against 3 real sovereign crises — Sri Lanka (2022 default) and Ghana (2022 debt restructuring), where scores showed multi-year deterioration ahead of each crisis, and Pakistan (2022–23 IMF emergency bailout), where scores continued deteriorating through the crisis window.
- Built 10 PostgreSQL analytical queries — using CTEs, window functions (LAG, LEAD, RANK), and multi-table joins across 3 normalised tables — spanning four risk pillars (solvency & liquidity, external sector & currency, fiscal quality & market perception, structural trajectory); identified and remediated a systemic DISTINCT ON/ORDER BY logic error that had been silently substituting IMF forecast values for observed actuals, correcting debt-to-GDP, misery-index, PPP fair-value, and CAGR calculations across affected economies.
- Delivered 6 interactive Tableau dashboards (Executive Summary, Solvency & Fiscal, External Sector & Trade, Risk Composition & Rankings, Model Validation) spanning 20+ worksheets and 150+ calculated fields, operationalising the Safe Score model with a diverging-scale risk map, green/amber/red status classification, parameter-driven country selection, and cross-filter actions for dynamic drill-down and investment decision support.
- Developed a Python pipeline (Pandas, NumPy) to automate multi-source data ingestion, preprocessing, and structured CSV generation feeding the PostgreSQL and Tableau layers (*in progress*).

Portfolio Optimisation & Financial Risk Management (*Excel*):

- Constructed an optimal 10-asset portfolio from a 50-asset universe using CAPM and Efficient Frontier analysis to evaluate risk-return trade-offs and identify best-fit allocation.
- Performed Value at Risk (VaR) and Monte Carlo stress testing under market-crash, liquidity-crisis, and bull-market scenarios to establish risk-adjusted breach thresholds for portfolio management.

PROFESSIONAL EXPERIENCE

Xanadu Realty

Bengaluru, India | Mar 2025 – Jun 2025

Management Trainee — Sourcing & Analytics

- Built an Excel-based lead mapping and deduplication solution across 5+ channel-partner networks, identifying 300+ high-intent prospects, improving data quality, and reducing estimated cost-per-qualified lead by ~25%.
- Developed an Excel dashboard for lead-to-walk-in performance tracking, enabling KPI reporting, funnel analysis, and data-driven resource allocation across acquisition channels.
- Performed campaign measurement, ROI analysis and trend analysis on lead-source performance, supporting conversion tracking, campaign optimisation, channel effectiveness, and budget prioritisation.

MCX India

Hyderabad, India | Apr 2024 – May 2024

Research & Analytics Intern — Financial Analytics

- Performed time-series and volatility analysis on 15+ years of MCX base-metals data, identifying market risk patterns and supporting commodity price forecasting for institutional investment analysis.
- Developed R-based visualisations to analyse commodity price trends, quantify downside risk, and support evidence-based investment and hedging decisions.
- Acknowledged for research on commodity price risk and hedging strategies, published on the MCX India website and reaching institutional investors internationally; assessed ESG and SDG compliance of global smelting firms to support governance, sustainability, and credit risk assessment.

TCL-POTPL

Tirupati, India | Aug 2019 – Mar 2022

Associate Process Testing Engineer & Team Lead

- Generated SOP-compliant MIS reports and process documentation while tracking production output and defect trends, improving reporting accuracy and strengthening audit readiness.
- Implemented an Excel-based inventory monitoring system that maintained optimal stock levels, eliminated production interruptions, and improved operational reporting accuracy.
- Led a cross-functional testing team to deliver a mass-production milestone 3 weeks ahead of schedule despite COVID-19 disruption, coordinating across 3 departments with zero critical defects at handover.

EDUCATION

MBA — Finance & Analytics

2023 – 2025

IBS Hyderabad, IFHE University | CGPA: 7.78/10

Relevant coursework: Business Analytics, Financial Modelling, Hypothesis Testing, Statistical Inference

B.Tech — Electronics & Communication Engineering

2015 – 2019

Andhra University College of Engineering | CGPA: 7.64/10

CERTIFICATIONS & ACHIEVEMENTS

- **Data Science Program — E&ICT IIT Guwahati** (*Expected: Sep 2026*) — Modules completed: SQL (PostgreSQL), Python for Data Analytics, Advanced Excel, Power BI | Currently pursuing Machine Learning and GenAI.
- **Published Research** — Acknowledged for research contribution to MCX-ICFAI Business School study, "Hedging of Price Risks in Base Metals," published on MCX India's Research Studies portal.
- **CAT 2022** - 92.93 percentile | **NMAT 2022-23** - 242 | National distinctions in AMTI, AIMEd, SIMO.

LANGUAGES

English — Professional | Hindi — Native | Telugu — Mother Tongue